ABG Motors Market Expansion Analysis Using Machine Learning

ABSTRACT

The objective of this project is to analyze customer purchasing behavior and identify potential customers for ABG Motors in the Indian market using Machine Learning techniques. The project uses customer datasets from the Japanese automobile market to build a predictive classification model.

Logistic Regression was used as the primary machine learning algorithm to predict whether a customer is likely to purchase a new vehicle based on demographic and financial attributes such as age, gender, annual income, and age of the current vehicle.

The trained model was then applied to the Indian customer dataset to identify potential customers in the Indian market. The analysis predicted 54,159 potential customers, indicating strong business expansion opportunities for ABG Motors.

Additionally, Tableau dashboards were developed to visualize customer demographics, purchasing behavior, income distribution, gender-based purchase trends, and Indian market predictions.

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1. INTRODUCTION

ABG Motors is planning to expand its automobile business into the Indian market. To support business decision-making, customer purchase behavior analysis is required to identify potential customers who are likely to purchase new vehicles.

Machine Learning techniques help organizations analyze customer data and predict future purchasing behavior using historical information. Predictive analytics allows businesses to optimize marketing strategies, identify target customers, and improve business profitability.

In this project, customer data from the Japanese market is used to train a machine learning model. The trained model is then used to predict potential customers in the Indian market.

2. PROBLEM STATEMENT

The main objective of this project is to develop a machine learning classification model that predicts whether a customer is likely to purchase a new vehicle based on demographic and financial attributes.

The model should:

- Analyze Japanese customer purchase data.
- Predict customer purchasing behavior.
- Identify potential customers in the Indian market.
- Provide business insights using Tableau visualizations.

3. OBJECTIVES

The objectives of the project are:

- To perform data preprocessing and cleaning.
- To analyze customer demographic and financial information.
- To build a classification model using Logistic Regression.
- To evaluate the performance of the machine learning model.
- To predict potential customers in the Indian market.
- To create interactive visualizations using Tableau.
- To provide business recommendations for ABG Motors.

4. DATASET DESCRIPTION

Two datasets were used in this project.

Japanese Dataset

The Japanese dataset contains customer information and purchase history.

Attributes:

Column Name	Description
ID	Customer ID
CURR_AGE	Current age of customer
GENDER	Gender of customer
ANN_INCOME	Annual income
AGE_CAR	Age of current vehicle
PURCHASE	Purchase status (0 or 1)

Dataset Size

- Total Records: 40,000

Indian Dataset

The Indian dataset contains customer demographic information.

Attributes:

Column Name	Description
ID	Customer ID
CURR_AGE	Current age
GENDER	Gender
ANN_INCOME	Annual income
DT_MAINT	Last maintenance date

Dataset Size

- Total Records: 70,000

5. TOOLS AND TECHNOLOGIES USED

Tool / Technology	Purpose
Python	Programming Language
Pandas	Data manipulation
NumPy	Numerical operations
Matplotlib	Data visualization
Seaborn	Statistical visualization
Scikit-learn	Machine learning
Tableau	Dashboard creation
VS Code	Development environment

6. METHODOLOGY

The project methodology includes the following steps:

1. Data Collection
2. Data Cleaning
3. Data Preprocessing
4. Exploratory Data Analysis
5. Feature Selection
6. Train-Test Split
7. Feature Scaling
8. Logistic Regression Model Training
9. Model Evaluation
10. Indian Market Prediction
11. Tableau Dashboard Creation

7. DATA PREPROCESSING

Data preprocessing is an important step in machine learning.

The following preprocessing techniques were applied:

Missing Value Checking

Both Japanese and Indian datasets were checked for missing values.

Result:

- No missing values were found.

Gender Encoding

The gender column was converted into numerical format:

Gender	Encoded Value
Male	1
Female	0

Feature Selection

The following features were selected for model training:

- Current Age
- Gender
- Annual Income
- Age of Car

Target Variable:

- PURCHASE

Feature Scaling

StandardScaler was used to normalize numerical values before model training.

8. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) was performed to understand customer purchasing behavior.

Customer Age Distribution

The customer age distribution analysis showed that customers aged between 30 and 60 years contributed significantly to the dataset.

Purchase Distribution

The analysis showed that the number of buyers was higher than non-buyers, indicating positive customer demand.

Gender vs Purchase Analysis

Both male and female customers demonstrated significant purchasing interest. Male customers showed slightly higher purchase activity.

Income Distribution Analysis

Higher income groups demonstrated stronger purchasing potential, indicating annual income significantly influences customer buying behavior.

9. MACHINE LEARNING MODEL

Logistic Regression

Logistic Regression is a supervised machine learning classification algorithm used to predict categorical outcomes.

In this project, Logistic Regression was used to predict whether a customer is likely to purchase a new vehicle.

Advantages of Logistic Regression

- Simple and efficient
- Suitable for binary classification
- Easy interpretation of coefficients
- Good performance for structured datasets

10. MODEL EVALUATION

The model was evaluated using multiple evaluation metrics.

Accuracy Score

Accuracy obtained:

68.41%

Classification Report

Metric	Class 0	Class 1
Precision	0.64	0.71
Recall	0.55	0.78
F1-Score	0.59	0.74

Confusion Matrix

Actual / Predicted	0	1
0	1833	1516
1	1011	3640

The model demonstrated satisfactory predictive performance.

11. BUSINESS INTERPRETATION OF COEFFICIENTS

The Logistic Regression coefficients provide insights into customer purchasing behavior.

Feature	Coefficient
Current Age	-0.126194
Gender	0.108943
Annual Income	0.403873
Age of Car	0.846388

Interpretation

- Annual income positively influences purchasing behavior.
- Customers with older vehicles are more likely to purchase new vehicles.
- Gender has a small positive impact.
- Current age has a slight negative relationship with purchasing probability.

12. INDIAN MARKET PREDICTION

The trained Logistic Regression model was applied to the Indian customer dataset.

Prediction Results

Category	Count
Potential Customers	54,159
Non-Potential Customers	15,841

The prediction results indicate strong business expansion opportunities for ABG Motors in the Indian market.

13. TABLEAU DASHBOARD ANALYSIS

A Tableau dashboard was developed to visualize customer and market trends.

Dashboard Components

- Customer Age Distribution Analysis
- Purchase Distribution Analysis
- Gender vs Purchase Analysis
- Income Distribution Analysis
- Indian Market Prediction Analysis

Dashboard Insights

- Buyer count is higher than non-buyers.
- Male customers show slightly higher purchasing activity.
- Higher income customers demonstrate stronger purchasing behavior.
- The Indian market shows significant expansion potential.

14. RESULTS AND FINDINGS

The project successfully developed a machine learning model capable of predicting customer purchase behavior.

Major Findings

- Logistic Regression achieved satisfactory prediction accuracy.
- Annual income strongly influences vehicle purchase behavior.
- Customers with older vehicles are more likely to purchase new vehicles.
- The Indian market contains a large number of potential customers.
- Tableau dashboards effectively visualize customer trends and business insights.

15. CONCLUSION

The project successfully implemented a machine learning-based customer purchase prediction system for ABG Motors.

The Logistic Regression model analyzed customer demographic and financial attributes to predict purchasing behavior. The trained model achieved satisfactory classification performance and identified 54,159 potential customers in the Indian market.

The Tableau dashboard provided clear business insights regarding customer demographics, purchasing behavior, income influence, and market opportunities.

The analysis indicates that the Indian market offers strong business expansion potential for ABG Motors.

16. FUTURE SCOPE

The project can be further improved in the future by:

- Using advanced machine learning algorithms such as Random Forest and XGBoost.
- Implementing deep learning techniques for higher accuracy.
- Developing a real-time customer prediction system.
- Integrating cloud-based analytics platforms.
- Enhancing dashboard interactivity and business intelligence reporting.

17. REFERENCES

1. Scikit-learn Documentation
2. Pandas Documentation
3. Tableau Documentation
4. Python Official Documentation
5. Machine Learning Research Articles

APPENDIX

Output Files Generated

- Indian_Market_Predictions.csv
- Model_Coefficients.csv
- Tableau Dashboard

Software Requirements

- Python 3.x
- Tableau Public/Desktop
- Visual Studio Code

END OF REPORT